

1 **Supporting Information**

2 **Intrinsic Defect Self-Compensation Enables Spectrum Narrowing of**
3 **Uniform Alloyed Ag-In-Ga-S Quantum Dots**

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Experimental Methods

Materials

Silver acetate (Ag(Ac), 99.99%, powder) was purchased from Adamas. Indium(III) iodide (InI₃, 99.999%, powder) and Gallium acetylacetonate (Ga(acac)₃, 99.99%, powder) were purchased from Aladdin and Macklin Inc, respectively. The sulphur (S, 99.99%, powder) and absolute ethanol (Analytical Reagent, AR) were purchased from Nanjing-reagent. Oleylamine (OAm, 90%), 1-octadecene (1-ODE, 90%), 1-dodecanethiol (1-DDT, 98%), and hexane (99.7%) were purchased from Aladdin. Toluene (AR) was purchased from Sinopharm Chemical Reagent Co., Ltd. Poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS), poly(bis(4-phenyl)(2,4,6-trimethylphenyl)amine) (PTAA), 1,3,5-Tris(1-phenyl-1H-benzimidazol-2-yl)benzene (TPBi) and LiF were brought from Xi'an Polymer Light Technology Corp.

Synthesis of AIGS QDs

Combining 0.1 mmol of Ag(Ac), 0.2 mmol of InI₃, and 0.05-0.3 mmol of Ga(acac)₃ with OAm (3.5 mL), 1-ODE (6.25 mL) and 1-DDT (1 mL) into a three-necked flask, then the three-necked flask was heated to 50°C and subjected to vacuum for 60-120 min, during which nitrogen (N₂) was filled for two times. And then, the temperature was adjusted to 150°C, the 1.25 mL of S precursor solution (0.64 mmol/mL) was rapidly injected once the set temperature was achieved. The reaction was maintained at 150°C for 10-20 min to get the broad-emitting AIGS QDs (BE-QDs). Thereafter, the reaction temperature was incrementally increased to 280 °C within 10-15 min to reactivate the unreacted precursors. After a reaction period of 30 min at 280°C, the narrow-emitting AIGS QDs (NE-QDs) were obtained as the reaction mixture cooled down to room temperature. The resulting crude QDs solution was subsequently transferred to 50 mL centrifuge tubes for further processing. The crude QDs solution was subjected to high-speed centrifugation of 8000 rpm to remove the precipitated by-products and collected the supernatant. Subsequently, twice the volume of absolute ethanol was introduced to the aforementioned solution, followed by another round of centrifugation (8000 rpm/5 min). The precipitate obtained from this centrifugation was then retrieved and redispersed in a toluene solution. Additional twice the volume of ethanol was thereafter introduced into the QDs-toluene dispersion. The precipitate was once more separated by centrifugation (8000 rpm/5 min) and finally redispersed in either hexane or toluene for subsequent applications.

Device Fabrication

The QLEDs structure consisting of indium tin oxide (ITO)/ poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS)/poly (bis(4-phenyl)(2,4,6-trimethylphenyl) amine) (PTAA)/QDs/1,3,5-Tris (1-phenyl-1H-benzimidazol-2-yl) benzene (TPBi)/LiF/Al.

PEDOT:PSS solutions (Baytron P VPAI 4083) were spin-coated onto the ITO-coated

1 glass substrates at 3000 rpm for 45 s and baked at 140°C for 15 min. PTAA (in
2 chlorobenzene, 5 mg mL⁻¹) and AIGS QDs (in toluene or hexane, 10-15 mg mL⁻¹) were
3 deposited layer by layer through spin coating at 2000 rpm for 60 s. PTAA layers was
4 baked at 120°C for 20 min. TPBi (40 nm) and LiF/Al electrodes (1 nm/100 nm) were
5 deposited by thermal evaporation in a vacuum indicator (2×10^{-4} Pa). The device active
6 area was 4 mm² as defined by the overlapping area of the ITO and Al electrodes.

7 **Materials characterizations and Device Measurements**

8 The absorbance was measured by a UV-VIS-NIR spectrophotometer (Shimadzu UV-
9 3600). Photoluminescence (PL) spectra of the QD solutions were analyzed using a
10 Varian Cary Eclipse spectrometer. X-ray diffraction (XRD) samples were prepared by
11 depositing purified QD solutions onto glass substrates. XRD patterns were collected
12 using a Bruker D8 Advance X-ray diffractometer with Cu K α radiation ($\lambda = 1.5406$ Å).
13 Transmission electron microscopy (TEM) images were obtained using a Tecnai G2 F30
14 S-TWIN TEM instrument at an acceleration voltage of 200 kV. Absolute
15 photoluminescence quantum yields (PLQYs) were determined using a Horiba
16 Fluorolog system equipped with a Quanta-Phil integration sphere. Time-resolved
17 photoluminescence (TRPL) decay lifetimes were measured using a time-correlated
18 single-photon counting (TCSPC) spectrofluorometer (FLS920, Edinburgh Instruments,
19 UK) at room temperature. Fourier Transform Infrared Spectroscopy (FTIR) was
20 conducted using a JASCO-FT/IR-4700 system. The surface chemical electronic states
21 and compositions of the QDs were examined using X-ray photoelectron spectroscopy
22 (XPS) with a monochromatic Al-K α radiation source (ESCALAB 250Xi spectrometer,
23 Thermo Scientific, USA). Elemental compositions were assessed using a NexION 300x
24 inductively coupled plasma mass spectrometer (ICP-MS). The performance of QLEDs
25 was evaluated using an LED testing system from Everfine Photo-E-Info Co., Ltd.,
26 employing a Keithley 2400 source meter and a Chroma Meter CS-2000. EPR data were
27 taken with a Bruker EMX spectrometer operating near 9.48 GHz.

1 The E_u values of QDs were derived using the following equation:

2
$$\alpha = \alpha_0 \exp(h\nu/E_u) \quad (S1)$$

3
$$\ln \alpha = \ln \alpha_0 + (h\nu/E_u) \quad (S2)$$

4 where, α_0 is a constant, α denotes the absorption coefficient, $h\nu$ represents photon
5 energy, and E_u is defined as the Urbach energy. Based on the equation S(1) and S(2),
6 E_u could be derived as the reciprocal of the slope.

7
8 The AIGS QDs TRPL spectra curves were fitted and analyzed by a tri-exponential
9 function shown as below:

10
$$I(t) = A_1 \exp(-t/\tau_1) + A_2 \exp(-t/\tau_2) + A_3 \exp(-t/\tau_3) \quad (S3)$$

11 The average lifetimes (τ_{ave}) of AIGS QDs were calculated by follow equation:

12
$$\tau_{ave} = (A_1 \tau_1^2 + A_2 \tau_2^2 + A_3 \tau_3^2) / (A_1 \tau_1 + A_2 \tau_2 + A_3 \tau_3) \quad (S4)$$

13 The detailed analysis of the occurrence ratio of each recombination mechanism was
14 calculated by the formula:

15
$$f_i = A_i \tau_i / (A_1 \tau_1 + A_2 \tau_2 + A_3 \tau_3) \quad (i=1, 2, 3) \quad (S5)$$

16 where τ_1 , τ_2 , and τ_3 represent the average lifetimes of different carrier recombination
17 processes, respectively. The amplitudes corresponding to each of these components are
18 denoted as A_1 , A_2 , and A_3 , respectively. f_1 , f_2 and f_3 represent the contribution
19 proportions of non-radiative recombination, band-edge radiative recombination, and
20 defect-related radiative recombination, respectively.

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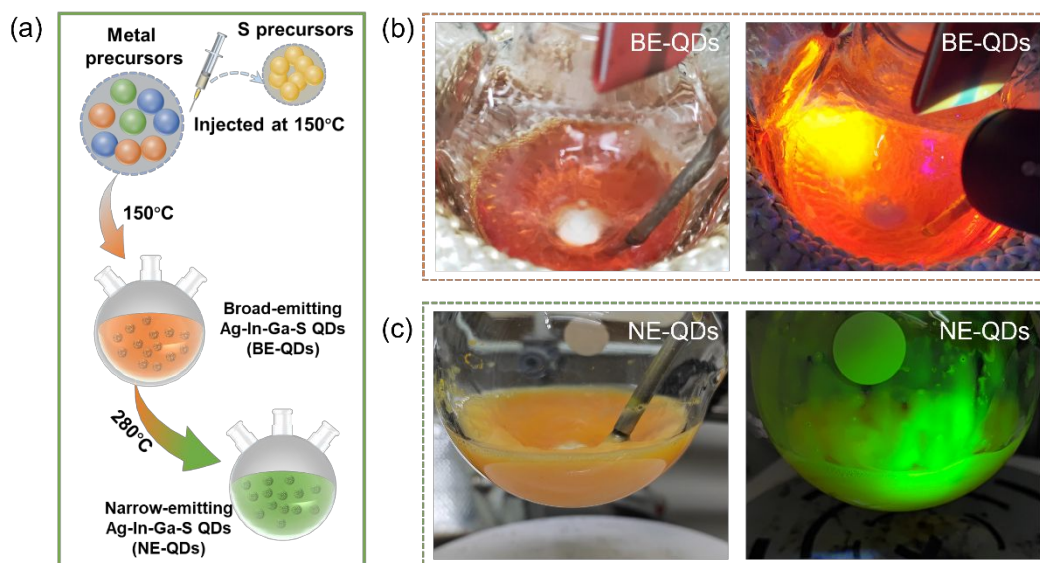


Figure S1. (a) Schematic diagram of the synthesis method for AIGS QDs. Photographs of AIGS QDs solution at (b) 150°C (BE-QDs) and (c) 280°C-30 min (NE-QDs) under sun light (left) and UV light (right).



Figure S2. Photograph of AIGS QDs solution in different reaction stages under UV light. As the temperature increased from 150°C to 280°C, the fluorescence brightness gradually decreased, which was attributed to the fluorescence thermal quenching effect. With the increase in reaction time, the fluorescence brightness of QDs gradually recovered, turning the color from orange to yellow and finally presenting a bright green fluorescence.

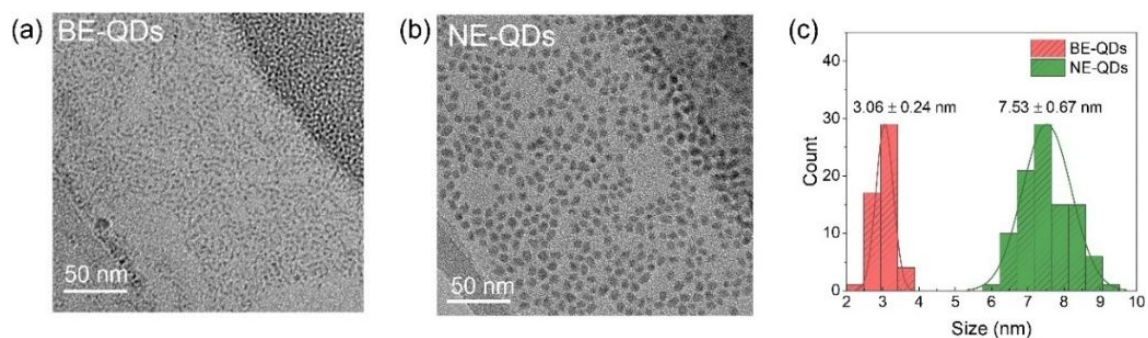


Figure S3. TEM images of (a) BE-QDs and (b) NE-QDs, respectively. (c) Statistical distribution of particle size of BE-QDs and NE-QDs.

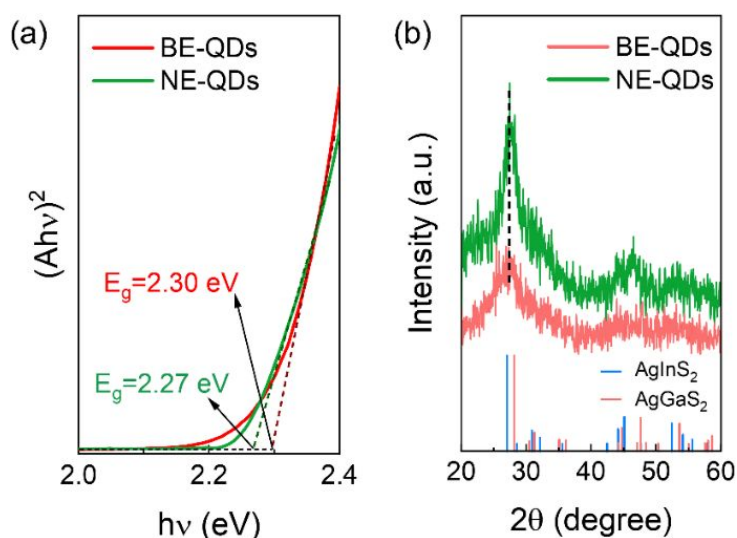


Figure S4. (a) The bandgaps of BE-QDs and NE-QDs are determined by the absorption spectrum. (b) XRD patterns of BE-QDs and NE-QDs and reference XRD patterns of AgInS₂ (PDF#75-0117) and AgGaS₂ (PDF#75-0114).

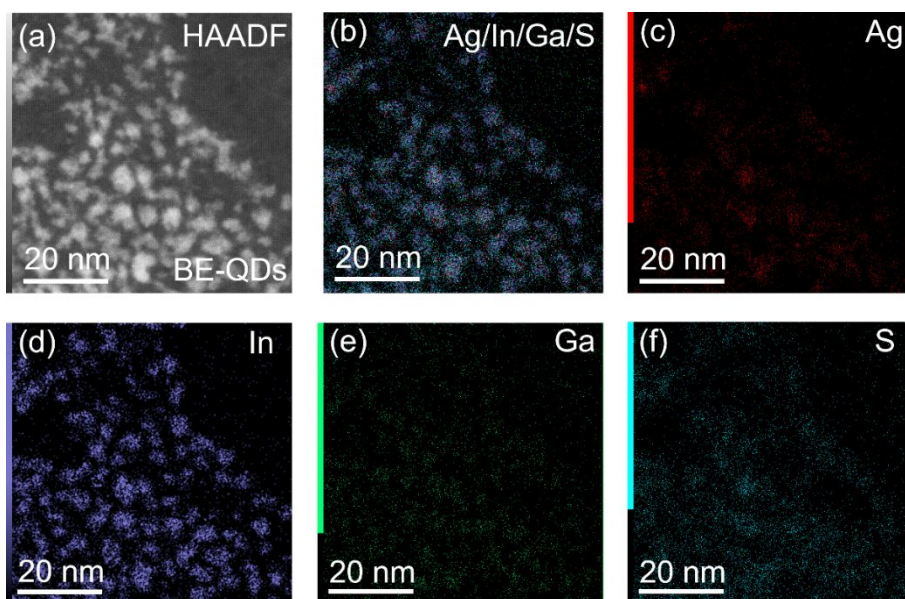


Figure S5. (a) STEM-HAADF image, and (b)- (f) EDX elemental mapping of BE-QDs.

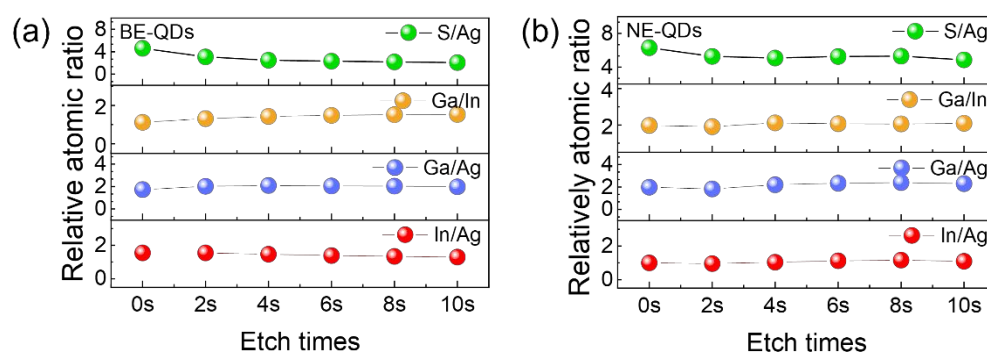


Figure S6. The relative atomic ratios of each element of (a) BE-QDs and (b) NE-QDs under different XPS etching times (etching depth per second: 0.2 nm).

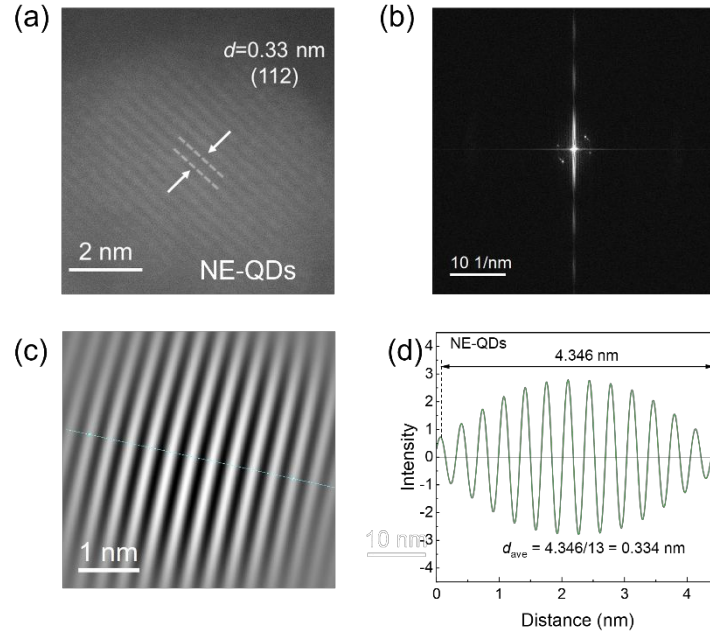


Figure S7. (a) STEM-HAADF image of a single NE-QD and (b) the corresponding Fast Fourier transform (FFT) and (c) inverse FFT (iFFT) of the whole area of a single NE-QD. (d) Line profile of iFFT.

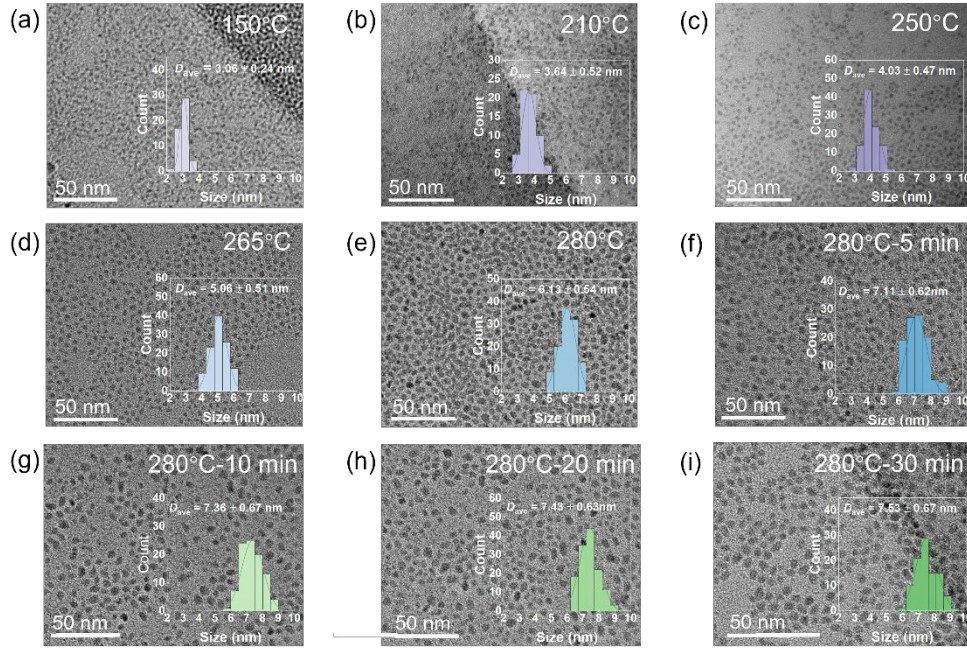


Figure S8. TEM images of AIGS QDs at (a) 150°C, (b) 210°C, (c) 250°C, (d) 265°C, (e) 280°C, (f) 280°C-5 min, (g) 280°C-10 min, (h) 280°C-20 min, and (i) 280°C-30 min, respectively.

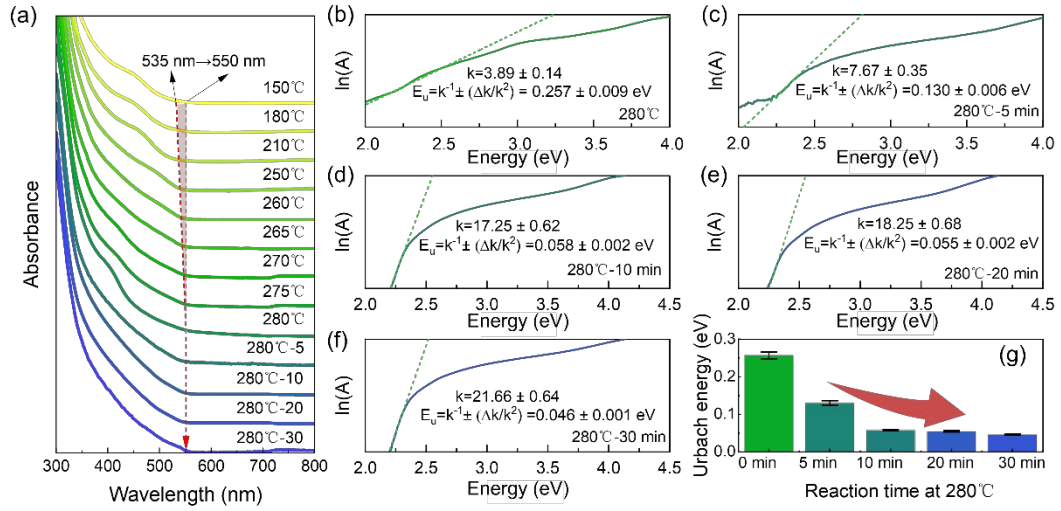


Figure S9. (a) UV-vis absorption spectra of AIGS QDs during the growth. Relationship of $\ln(A)$ versus energy for the calculation of Urbach energy at (b) 280°C, (c) 280°C-5 min, (d) 280°C-10 min, (e) 280°C-20 min, and (f) 280°C-30 min, respectively. (g) Urbach energy of AIGS QDs at 280°C reaction stage calculated by UV-vis absorption spectra.

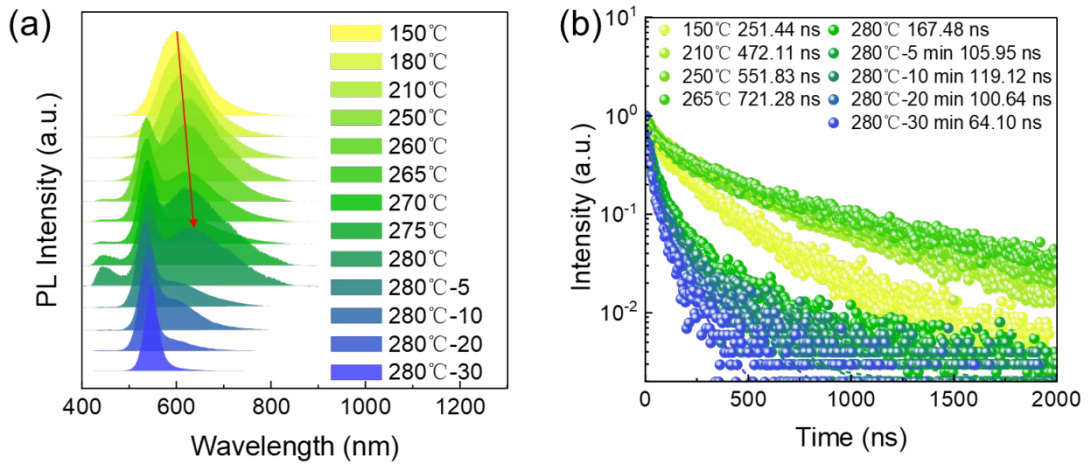


Figure S10. (a) PL spectra and (b) TRPL spectra of AIGS QDs growth evolution.

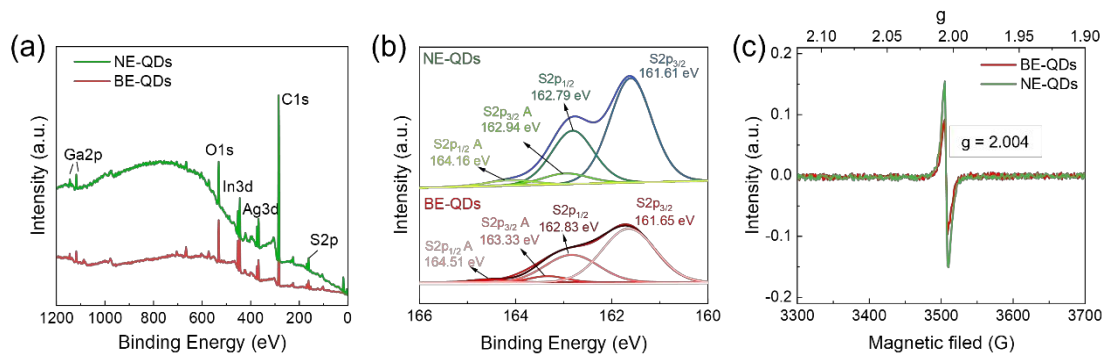


Figure S11. (a) The XPS spectra of BE-QDs and NE-QDs. The positions of orbital peak binding energies of all metal elements demonstrated the presence of Ag^+ , In^{3+} and Ga^{3+} , and S^{2-} . (b) High resolution XPS spectra of S2p. (c) EPR spectra of BE-QDs and NE-QDs.

Considering that the reduction of sulfur vacancies (V_s) might play a role in narrow emission, the changes in V_s concentration were investigated and analyzed using electron paramagnetic resonance (EPR). As shown in Figure S11c, the EPR signal ($g = 2.004$) confirmed the existence of sulfur vacancy (V_s),¹ and a stronger signal intensity indicated that the concentration of V_s in NE-QDs was higher than that in BE-QDs. The decrease of S2p orbital binding energy in XPS (Figure S11b) also confirmed the increase of V_s .² The lower defect-related radiative recombination contribution proportion (f_2) in NE-QDs (Table S2) indicated that the change in V_s did not have a positive effect on the emission narrowing. Thus, the realized of narrow emission in AIGS QDs is independent of changes in V_s defects.

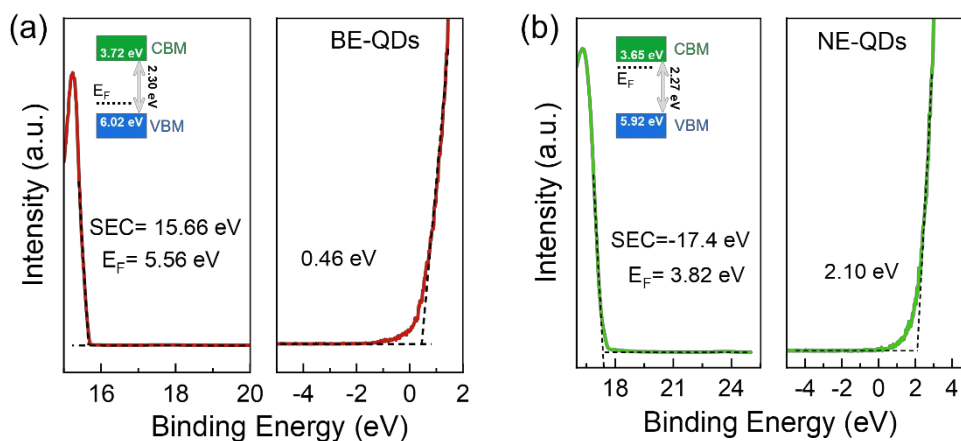


Figure S12. UPS spectra of (a) BE-QDs and (b) NE-QDs, respectively. Inset, schematic diagram of energy level structure. The positions of the secondary electron cutoff edge (SEC) and the distance from the Fermi level (E_F) to the VBM (E_F -VBM) verified the positions of the E_F of the two types of QDs.

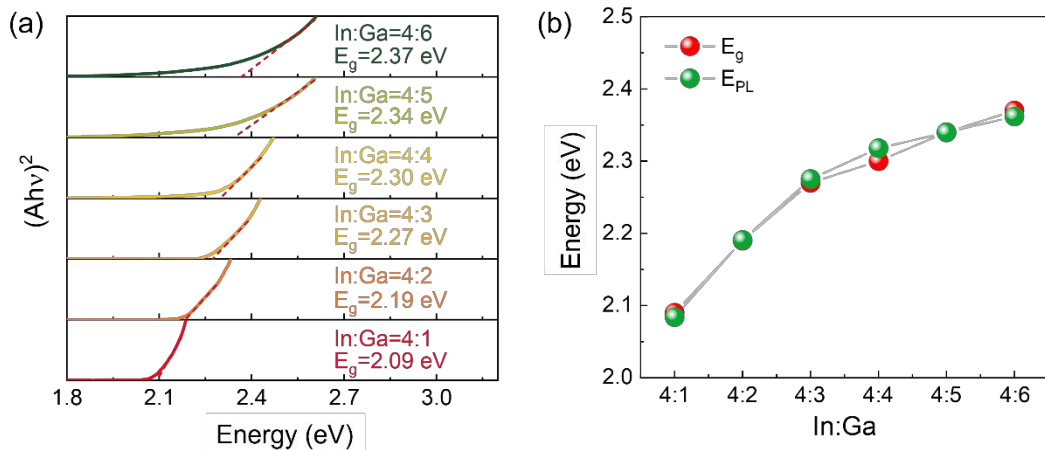


Figure S13. (a) The bandgaps (E_g) of NE-QDs with varying In:Ga ratios are determined by the absorption spectra, (b) comparison between E_g and E_{PL} of NE-QDs with varying In:Ga ratios.

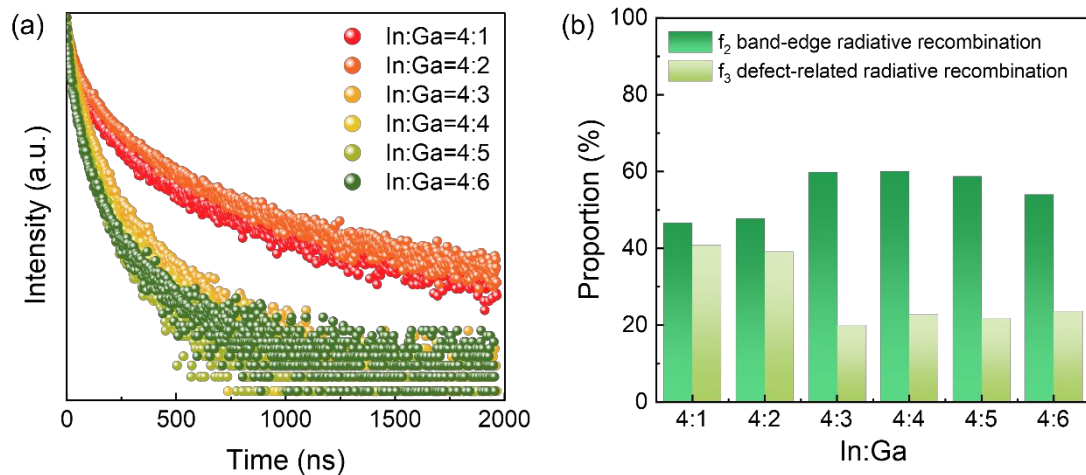


Figure S14. (a) The TRPL spectra curves of NE-QDs with different In:Ga, (b) proportion of different recombination mechanism of NE-QDs with different In:Ga.

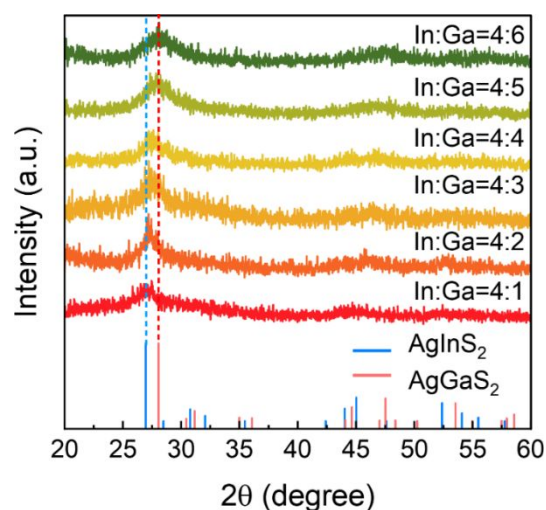


Figure S15. The XRD spectra of NE-QDs with different In:Ga.

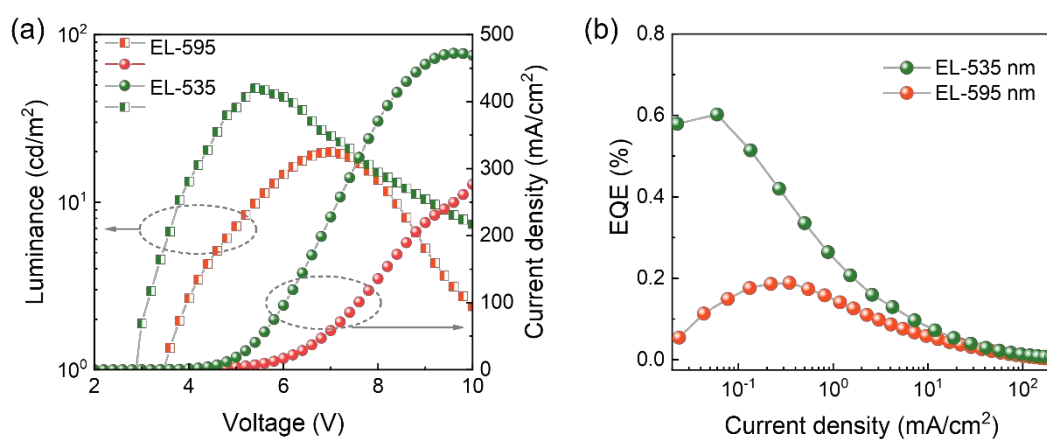


Figure S16. Electrical performance of QLEDs based on narrow-emitting AIGS QDs. (a) Voltage–luminance (V–L, square dot lines) and current density–voltage (J–V, spherical dot lines) characteristics, (b) EQE versus current density.

Table S1. The relative atomic ratios of each element of BE-QDs and NE-QDs under different XPS etching times (etching depth per second: 0.2 nm).

Sample	Etch times/s	In/Ag	Ga/Ag	(In+Ga)/Ag	Ga/In	S/Ag
BE-QDs	0	1.55	1.73	3.28	1.12	4.62
	1	1.55	2.02	3.57	1.31	3.08
	2	1.47	2.09	3.56	1.43	2.49
	3	1.39	2.05	3.44	1.48	2.32
	4	1.34	2.04	3.38	1.52	2.19
	5	1.30	1.98	3.28	1.53	2.06
NE-QDs	0	1.00	1.98	2.98	1.99	6.30
	2	0.96	1.83	2.79	1.91	5.28
	4	1.04	2.20	3.24	2.12	5.08
	6	1.13	2.33	3.36	2.07	5.26
	8	1.16	2.39	3.55	2.06	5.30
	10	1.10	2.30	3.40	2.09	4.87

Table S2. Fitted parameter of time-resolved photoluminescence (TRPL) spectra of AIGS QDs at different reaction stage.

Sample	τ_{ave}/ns	τ_1/ns	τ_2/ns	τ_3/ns	A_1	A_2	A_3	$f_1/\%$	$f_2/\%$	$f_3/\%$
150°C	251.44	9.77	78.49	292.11	0.23	0.32	0.41	1.51	17.04	81.45
210°C	472.11	29.73	202.22	643.61	0.28	0.46	0.26	3.09	34.56	62.35
250°C	551.83	9.06	121.84	658.95	0.24	0.41	0.32	0.84	18.93	80.23
265°C	721.28	34.52	245.34	970.59	0.35	0.39	0.22	3.78	29.49	66.72
280°C-0 min	167.48	5.94	45.94	268.85	0.36	0.52	0.12	3.66	41.16	55.18
280°C-5 min	105.95	3.75	27.73	168.72	0.27	0.60	0.13	2.49	41.61	55.90
280°C-10 min	119.12	4.86	30.37	194.96	0.27	0.61	0.12	3.07	42.53	54.40
280°C-20 min	100.64	7.29	35.80	198.98	0.38	0.54	0.08	7.47	51.49	41.04
280°C-30 min	64.10	6.00	29.51	129.80	0.34	0.58	0.08	6.94	56.94	36.12

Table S3. Atomic ratios of BE-QDs and NE-QDs calculated by ICP-MS.

Sample	Ag/%	In/%	Ga/%	Ag:(In+Ga)/%
BE-QDs	25.36	55.83	18.81	33.98
NE-QDs	27.30	33.36	39.34	37.55

Table S4. Atomic ratios of BE-QDs and NE-QDs calculated by XPS.

Sample	Ag/%	In/%	Ga/%	S/%	(M-S)/%	(-SH)/%	Ag:(In+Ga)/%
BE-QDs	12.74	20.99	10.36	55.91	53.25	2.66	40.64
NE-QDs	13.98	16.87	10.25	58.90	55.07	3.83	51.55

Table S5. Synthesis and performance summary of narrow-emission AIGS QDs in recent years.

QDs	Synthesis	PLQY/%	FWHM/nm	Year
AIGS/GaS _x ³	Heat-up	28-59	32-42	2021
AIGS/AGS ⁴	Hot injection	49-96	30-55	2023
AIGS/GaS _x ⁵	Hot injection	20-75	31-37	2023
AIGS ⁶	Heat-up	16	40	2024
AIGS/AGS/GaS ₂ S ₃ ⁷	Hot injection	95	33	2024
AIGS ⁸	Seed-mediated growth	/	38	2024
AIGS/GaS _x ⁹	Heat-up	90	31	2024
Zn-AIGS ¹⁰	Heat-up	14-45	34-48	2024
AIGS	Hot injection	78-96.1	30-38	This work

Table S6. Fitted parameter of time-resolved photoluminescence (TRPL) spectra of NE-QDs with different In:Ga.

Sample	τ_{ave}/ns	τ_1/ns	τ_2/ns	τ_3/ns	A ₁	A ₂	A ₃	f ₁ /%	f ₂ /%	f ₃ /%
In:Ga=4:1	322.81	17.30	120.15	647.13	2911.21	1582.74	256.93	12.38	46.75	40.87
In:Ga=4:2	355.81	25.97	154.86	710.33	2741.26	1684.82	301.45	13.03	47.77	39.20
In:Ga=4:3	101.44	16.11	63.88	287.78	2790.54	2180.77	169.13	19.30	59.81	20.09
In:Ga=4:4	78.07	10.24	46.97	211.83	2998.9	2279.5	191.02	17.23	60.07	22.70
In:Ga=4:5	67.13	9.42	43.45	183.45	3228.04	2134.3	187.73	19.30	58.85	21.85
In:Ga=4:6	74.92	8.96	43.70	208.04	3587.56	1790.28	164.74	22.22	54.09	23.69

Table S7. Performance summary of QLEDs based on narrow-emission AIGS QDs in recent years.

QDs	Structure	EL/nm	FWHM/nm	EQE/%	Year
AIGS/GaS _x ¹¹	core/shell	531	33	1.1	2023
Zn-AIGS ¹⁰	alloyed	470	48	0.53	2024
Zn-AIGS ¹⁰	alloyed	525	34	0.13	2024
Zn-AIGS ¹⁰	alloyed	614	38	0.03	2024
AIGS	alloyed	535	35	0.60	This work
AIGS	alloyed	595	33	0.19	

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